WEB APPENDIX

Stockpiling Points in Linear Loyalty Program

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Web Appendix: Stockpiling Points in Linear Loyalty Programs

Part 1: a rational expectations alternative

This web appendix empirically explores the directional predictions of a Rational Expectations (RE) model, an enriched SLA model in which consumers are forward-looking with rational expectations on the timing and size of their future shopping needs. A customer in this model views points as a means to save cash in the future, and does not inherently value points in a separate account. Behavior can be driven by foregone points and fixed non-monetary costs, but not separate mental accounts. For simplicity, we omit the i subscripts when presenting the individual-level model.

Consider a consumer who is forward looking over an infinite horizon and who values monetary incentives with a linear utility, as shown in Equation 1. As before, when the customer redeems, he pays $-m + \tilde{s}$ instead of $-m$, and can also incur a non-monetary cost of redeeming c.

$$u(y|s,m) = \begin{cases} -\beta_m m & \text{if } y = 0 \\ -\beta_m(m - \tilde{s}) - c & \text{if } y = 1 \end{cases} \tag{1}$$

As in MA, this consumer understands that redeeming affects his future stockpile s' (Equation 2). However, he does not value all future points equally. The value for s' depends on *how* and *when* he expects to use them. We assume rational expectations on future shopping needs m' and future inter-purchase times. Let q be the distribution of m' conditional on the current basket price m. Future payoffs are discounted by β, interpreted as the daily discount rate δ times the expected duration until the next purchase. The consumer optimizes his redemptions to maximize expected total discounted utility (Equation 3).

$$s' = \begin{cases} s + mr & \text{if } y = 0 \\ s - \tilde{s} & \text{if } y = 1 \end{cases} \tag{2}$$

$$V(s,m) = \max_{y \in \{0,1\}} u(y|s,m) + \beta E[V(s',m')|s,m,y] \tag{3}$$

We use Chebyshev regression (Judd 1998), a flexible non-parametric approach, to approximate the value function $V(s, m_{level})$ at each of the two price levels: *low* and *high* (see part 2 of the online appendix for details). Discretizing m allows us to convey the intuition behind the basic tradeoffs that determine how redemptions vary with shopping needs. The estimates are used to calculate $z_{RE}(s, m_{level})$, the difference between the utility of redeeming and not redeeming in the current period (Equation 4). The consumer redeems when $z_{RE}(s,m) > 0$. Let s^* be the stockpile level at which $z_{RE}(s^*, m)$=0.

$$z_{RE}(s,m) = \beta_s \tilde{s} - c + \beta[E[V(s',m')|s,m,1] - E[V(s',m')|s,m,0]] \tag{4}$$

We compare $z_{RE}(s^*, m_{low})$ and $z_{RE}(s^*, m_{high})$ to intuitively describe the model's directional predictions under two cases: when c is 0 and when c is large. When $c = 0$, foregone points drive stockpiling behavior. Since foregone points are proportional to m, a consumer will prefer to redeem on a small price rather than a high price. In other words, s^* increases with m (i.e., $z_{RE}(s^*, m_{low}) < z_{RE}(s^*, m_{high})$). When c is substantial, both the economic and cognitive motivations are allowed to drive behavior. In this case, s^* instead decreases with m. Intuitively, he stockpiles longer to save more at each redemption. A large c may seem necessary ex-ante to explain the observed persistent stockpiling because (i) foregone points are small in magnitude (only 1% of the basket price) and (ii) individuals often neglect to consider the absence of potential positive cash flows (Frederick et al. 2009).

To recap, RE expects redemptions to decrease with m if c is small, and to instead increase with m if c is substantial. To explore whether either of these predictions are directionally consistent with our data, we examine the individual-level estimates from SLA, a model linear in $\tilde{s}$ and m. At the individual-level, β_{si} is positive for 100% of customers (with a 95% posterior interval ranging from 0.022 to 0.250 across individuals), and β_{mi} is negative

for most customers (95.7%) and not different from zero for the rest with a 95% posterior interval ranging from -0.151 to 0.003 across individuals). So, to be consistent with this data, RE implies that foregone points (the economic motivation) must be the main determinant of persistent stockpiling and *not* cognitive fixed costs. In contrast, MA can capture this empirical pattern regardless of whether a customer considers foregone points (to see why, refer back to Table 5 and set $a_{vi} = 0$).

A second exploratory analysis did not find preliminary evidence for RE in the data. We ran a logistic regression (Equation 5) of redemptions with the observed running means of each individual's shopping needs $\bar{m}_{ij}$ (i.e., $\bar{m}_{ij} = \frac{1}{j}\sum_{k=1}^{j} m_{ik}$) and of the days between purchases $\bar{d}_{ij}$, two factors that should influence consumers with rational expectations above and beyond $\tilde{s}_{ij}$ and m_{ij}. Fixed effects α_i were added to control for differences across individuals[1].

$$logit(p_{ij}) \sim \alpha_0 + \alpha_i + \beta_s \tilde{s}_{ij} + \beta_m m_{ij} + \gamma_m \bar{m}_{ij} + \gamma_d \bar{d}_{ij} \tag{5}$$

While RE with low c expects that redemptions decrease with expected shopping needs ($\gamma_m < 0$) and increase with expected durations ($\gamma_d > 0$), both of these parameters were positive and not significantly different from zero (p-values 0.50 and 0.99 respectively). The control variables were consistent with the results from SLA: β_s is positive ($4.5 * 10^{-3}$, p-value < 0.001) and β_m is negative ($-5.2 * 10^{-4}$, p-value $= 0.019$) .

In summary, to be consistent with this dataset, RE implies that persistent stockpiling must be mostly driven by the economic incentive of foregone points and *not* by cognitive costs. In contrast (as discussed in the paper), MA finds that differences in redemption behavior across individuals are mostly driven by cognitive and psychological motivations to redeem (fixed costs and separate accounts). While we do not rule out either model, the common neglect of foregone gains (Frederick et al. 2009) brings in question the plausibility

[1]The advantage of using running means rather than the mean across the entire individual's observed purchases is that it allows us to control for individual-level fixed effects. This helps overcome the difficulty of disentangling model differences from heterogeneity (e.g., customers with longer durations may have systematically different discount rates, and the distributions of shopping needs may tend to vary with the value for cash across individuals).

of RE for this dataset.

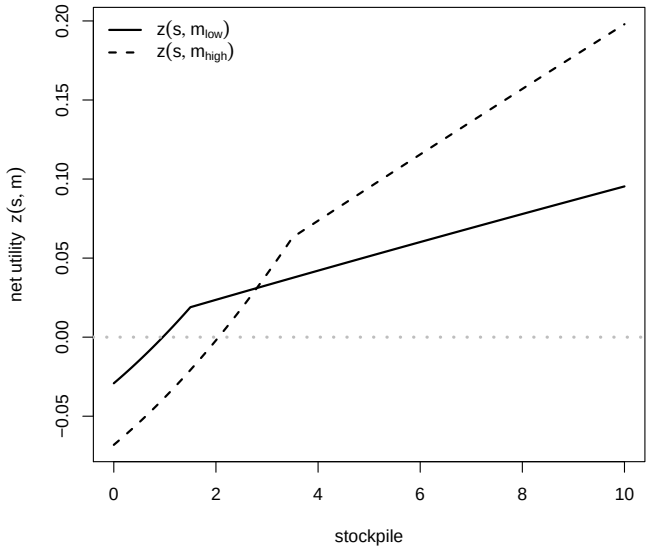
Part 2: value function estimation

The value function is approximated using orthogonal polynomials on 5 Chebyshev nodes (i.e., the zeros of a Chebyshev polynomial) over the stockpile levels as: $\tilde{V}(s, m_{level}) = \sum_{k=1}^{n} c_{k,level} T_k(s)$, where $T_k(s)$ denotes the Chebyshev basis function of order k, and $c_{k,level}$ are the coefficients. The $2n$ coefficients are estimated by numerically solving the system of $2n$ equations: $0 = \tilde{V}(s, m_{level}) - V(s, m_{level})$.

In the example shown in Figure A1, we used the following example parameters: $c = 0$, $\beta_m = 1$, $\beta = 0.98$, $m_{low} = 3$, $m_{high} = 7$, $q(m_{low}|m_{low}) = 0.2$, and $q(m_{high}|m_{high}) = 0.9$. The intuitive results do not change when the transition matrix q is modified so that m_{low} is more common than m_{high}. The kinks at the maximum redeemable amount ($m\kappa$) reflect that redeemable points at any purchase occasion are bounded at 50% of the basket price. The consumer persistently stockpiles up to the level s^* at which $z_{RE}(s^*, m)$=0.

When $c = 0$, the figure shows that s^* increases with m (i.e., $z_{RE}(s^*, m_{low}) < z_{RE}(s^*, m_{high})$). The same intuition holds when the transition matrix q is modified so that m_{low} is more common than m_{high}. When cognitive costs c are substantial, the consumer stockpiles longer and redeems on basket prices that are high instead of low. To see this graphically, refer again to Figure A1. As c increases, both curves move down. When c is large enough so that the zero crossing point occurs once $z_{RE}(s, m_{high})$ has surpassed $z_{RE}(s, m_{low})$, the customer's threshold s^* decreases with m.

Figure A1: Net utilities when shopping needs are low and high



References

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